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Inverse distance weighting interpolation-based immersed boundary velocity correction method for incompressible flows

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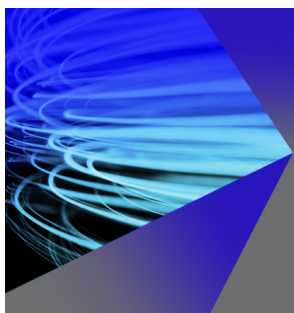
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ABSTRACT

In this work, the inverse distance weighting (IDW) interpolation is introduced into the implicit velocity correction-based immersed boundary method (IBM) for simulation of incompressible flows. In the original implicit velocity correction-based IBM, the solid body must be immersed in a uniform mesh region due to the use of the smooth Dirac delta function, which is utilized to associate Lagrangian points with their surrounding Eulerian points and only works with uniform meshes. The IDW method has the advantage that the interpolation range can be set flexibly. The introduction of the IDW interpolation can extend the application of the IBM to non-uniform meshes while reducing the number of Lagrangian points. The numerical test by the decaying vortex problem proves that the IDW interpolation does not significantly affect the overall accuracy of the IBM. In addition, numerical experiments for the flows around a circular cylinder and a NACA0012 airfoil demonstrate the advantages of the proposed method, including allowing fewer Lagrangian points while ensuring no streamline penetration to the solid body, as well as its adaptability to non-uniform meshes which can improve the computational efficiency due to the use of fewer mesh points. Finally, the simulation of the flow past a stationary sphere illustrates that the proposed method can effectively simulate the three-dimensional flow.

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I. INTRODUCTION

In computational fluid dynamics (CFD), most of the flow problems can be simulated by the body-fitted mesh method. However, for complex geometry and moving boundary problems, conventional body-fitted methods will encounter the difficulty of complex and tedious mesh generation. This difficulty can be overcome by the immersed boundary method (IBM). Basically, the IBM can be classified into the diffuse-interface one and the sharp-interface one. The diffuse-interface IBM allows smooth change of flow variables across the boundary, while the sharp-interface IBM considers sharp change of flow variables across the boundary. Examples of sharp-interface IBM include the ghost-cell method, the cut-cell method, the ghost-fluid

method, and so on. The diffuse-interface IBM is usually referred to as IBM. The first diffuse-interface IBM was proposed by Peskin,¹ which has become a popular approach for simulation of flows with complex geometry and moving boundary, due to its simplicity and easy implementation. The IBM uses a series of Lagrangian points to represent the solid boundary immersed in the fluid, and a fixed Eulerian mesh for the flow field. Then, the effect of the solid boundary on the flow field is considered by adding a forcing term to the governing equations. The determination of the forcing term, which reflects the interaction between the fluid flow and the immersed boundary, is a key issue for IBM. Since Peskin¹ introduced the original IBM, numerous modifications and improvements have been made. These variants are different

in ways to calculate the forcing term, such as the penalty forcing method,² feedback forcing method,³ physical virtual model,⁴ direct-forcing method,^{5,6} and so on.

In the conventional IBM proposed by Peskin,¹ the no-slip condition is not directly enforced so that some streamlines may penetrate the boundary surface at the converged state. Shu *et al.*⁷ thought that the unsatisfying of the no-slip boundary condition is because of the pre-calculated body force. By using the fractional step technique, they concluded that adding a body force in the momentum equations is equivalent to correcting the velocity field. In their approach, the velocity correction in the vicinity of the boundary point is made to enforce the boundary condition directly. So the no-slip boundary condition can be accurately satisfied. During the process of correcting velocity, they used linear interpolation along mesh lines. Following the idea of velocity correction, Wu and Shu⁸ developed the implicit velocity correction IBM. In their work, the velocity corrections at all boundary points are considered as unknowns, and the way to calculate them is that the velocity at the boundary interpolated from the corrected velocity field must satisfy the no-slip boundary condition. In order to improve accuracy and avoid oscillations at the boundary, the smooth Dirac delta function is adopted instead of the linear interpolation used in the work of Shu *et al.*⁷ Due to the simplicity of this approach, it has been widely used in various laminar flow problems.^{9–15} Based on Wu and Shu,⁸ Dash *et al.*^{16,17} proposed a flexible forcing method in which an additional sub-iteration update scheme is imposed to satisfy the no-slip condition within a convergence limit.

Since the Lagrangian points do not coincide with Eulerian mesh points, the interaction between fluid points and boundary points needs to use a suitable interpolation method. The smooth Dirac delta function^{18–20} is the most popular one used in the diffuse-interface IBM. It is indicated by Peskin¹⁸ that the Dirac delta function is smoothly approximated by a continuous kernel distribution, such as the 4-point piecewise function and the 4-point cosine function. Subsequently, Bao *et al.*¹⁹ presented a novel 6-point kernel, which yields a substantially improved translational invariance compared to other common IB kernels. Moreover, Yang *et al.*²⁰ developed a smoothing technique for discrete delta functions to suppress the nonphysical oscillations of volume forces in moving boundary problems. Their smoothed discrete delta functions were derived from the regular ones and were endowed with the higher order derivative compared to the original regular functions. However, the smooth Dirac delta function can only be applied to uniform Cartesian grids. What is more, due to the limitation of the influence range of the smooth Dirac delta function interpolation, the ratio between the Lagrangian point's spacing and the Eulerian mesh's spacing must be in a suitable range. Practically, there is a demand to introduce a more flexible interpolation function into the diffuse-interface IBM which is able to allow using fewer Lagrangian points at the same Eulerian mesh spacing. Reducing the number of Lagrangian points could improve the computational efficiency. In addition, it is hoped that the new interpolation method can allow using the non-uniform distribution of Lagrangian points as this is demanded in some practical problems. An example is the Lagrangian points distribution on the airfoil, where more points are distributed near the leading and trailing edges than in other regions.

The inverse distance weighting (IDW) method is a typical interpolation method with high efficiency and simplicity. It was originally proposed by Shepard²¹ to meet the needs in fields such as meteorology,

geography, and biology for interpolation from irregularly spaced data to produce a continuous surface. There are also some applications of IDW in CFD. Yang and Wu²² introduced the IDW to the ghost-cell (sharp-interface) IBM to simulate compressible flows with complex boundaries. In their work, boundary conditions are reconstructed by a serial of linear interpolation and IDW interpolation of flow variables in the fluid domain. Luo *et al.*²³ compared the IDW and bilinear interpolation schemes for the mirror points in their ghost-cell compressible IBM. Brahmachary *et al.*²⁴ used the IDW interpolation to reconstruct the solutions in the vicinity of the solid-fluid interface in their sharp-interface IBM for simulations of high-speed inviscid compressible flows. They showed that the proposed reconstruction has the second-order accuracy in space. Bharadwaj and Ghosh²⁵ compared three interpolation techniques based on IDW for data reconstruction. They used IDW to reconstruct pressure and shear stress at the immersed surface. In addition to the application for laminar flows, some researchers^{26–29} also applied the IDW combined with the sharp-interface IBM to simulate turbulent flows. Furthermore, Franke³⁰ proposed an alternative formulation of IDW, which has the property of preserving local maxima and producing smooth reconstruction. Gao *et al.*³¹ used this IDW interpolation to evaluate derivatives at the interpolated points in their ghost-point IBM. In addition, this scheme was also widely used in the sharp-interface IBM to get the value at the interpolated points from surrounding fluid points for simulation of turbulent flows.^{32–35} To sum up, the IDW is mainly applied in the sharp-interface IBM, in which the grid near the immersed boundary is classified into the fluid cell, solid cell and boundary cell such that the interpolation area of the interpolated points is always irregular. In this situation, the smooth Dirac delta function is no longer appropriate, because its interpolation area should be square and needs to be divided into x-, y-, and z-directions to calculate the interpolation coefficients. Different from it, the interpolation area of the IDW method is more flexible and the calculation of IDW coefficients is only related to the distance between the points.

Due to the flexibility and simplicity of the IDW, in the present work, we introduce the IDW interpolation into the diffuse-interface IBM. After comparing the coefficient distribution of the smooth Dirac delta function and the IDW function, we suggest a suitable power of inverse distance and the interpolation range of the IDW method. Some numerical experiments were performed to validate the present method. The example of the flow around a cylinder shows that our method can guarantee no streamline penetration to the cylindrical surface with fewer Lagrangian points as compared with the original diffuse-interface IBM, and in the meantime, predict the total force accurately. The numerical test by the decaying vortex problem proves that the IDW does not significantly affect the computational accuracy of the IBM. In addition, the simulation of the flow over a NACA0012 airfoil demonstrates the adaptability of the present method for application on the non-uniform mesh. Finally, the example of the flow past a stationary sphere illustrates the capability of the present method for the three-dimensional simulation.

II. IMPLICIT VELOCITY CORRECTION-BASED IMMERSED BOUNDARY METHOD

A. Governing equations

In IBM, as shown in Fig. 1, the flow field is represented by a fixed Eulerian mesh, and for the solid boundary immersed in the fluid, it is denoted by a series of Lagrangian points. Then, the effect of the

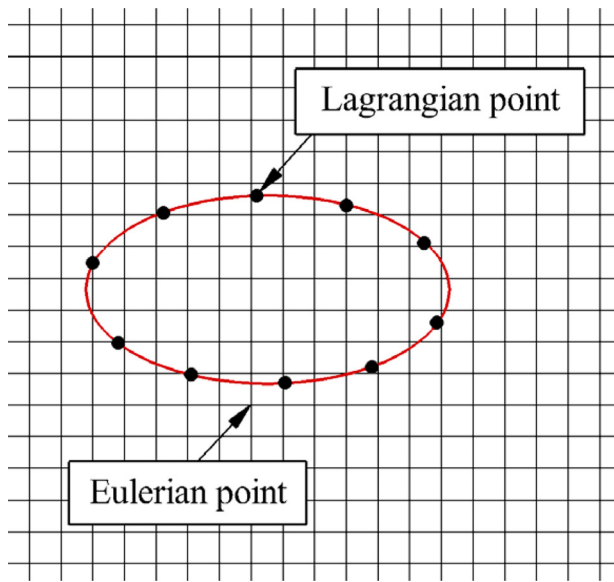


FIG. 1. Immersed boundary illustration. The black solid points are used to denote the Lagrangian points and the Eulerian mesh are represented by the Cartesian mesh.

immersed boundary on the flow field is considered by adding the forcing term into the momentum equation.

For incompressible flows, the governing equations can be written in a weakly compressible form as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (1)$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot [\mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] + \mathbf{f}. \quad (2)$$

Here ρ , p , and $\mathbf{u}$ are, respectively, the density, pressure, and velocity vector. $\mathbf{f}$ denotes the force density acting on the fluid, and μ is the dynamic viscosity.

B. Correction of the velocity field

In the implicit velocity correction-based IBM,⁸ the governing equations (1) and (2) are solved by using the fractional step technique, that is, the prediction step and the correction step:

Prediction step: Solve the N-S equations without the body force ($\mathbf{f} = 0$) to obtain the predicted velocity field $\mathbf{u}^*$.

Correction step: Correct the velocity field by solving

$$\frac{\partial \rho}{\partial t} = 0, \quad (3)$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} = \mathbf{f}. \quad (4)$$

From Eqs. (3) and (4), we can obtain

$$\rho \frac{\partial \mathbf{u}}{\partial t} = \mathbf{f}. \quad (5)$$

Equation (5) is discretized by

$$\rho \delta \mathbf{u} = \mathbf{f} \delta t, \quad (6)$$

where $\delta \mathbf{u}$ is the velocity correction and δt is the time size. Then, we can get the corrected velocity

$$\mathbf{u}^{l+1} = \mathbf{u}^* + \delta \mathbf{u} = \mathbf{u}^* + \frac{1}{\rho} \mathbf{f} \delta t, \quad (7)$$

where $\mathbf{u}^{l+1}$ represent the corrected velocity in the next time step.

The velocity at the boundary (Lagrangian points) can be interpolated from those of the surrounding fluid points (Eulerian points), i.e.,

$$\mathbf{U}^{l+1}(\mathbf{X}_m) = \sum_n \mathbf{u}^{l+1}(\mathbf{x}_n) D(\mathbf{x}_n - \mathbf{X}_m) h^3, \quad (8)$$

$(m = 1, 2, \dots, M; n = 1, 2, \dots, N),$

where $\mathbf{U}^{l+1}(\mathbf{X}_m)$ denotes the velocity at the Lagrangian point $\mathbf{X}_m = (X_m, Y_m, Z_m)$, $\mathbf{u}^{l+1}(\mathbf{x}_n)$ represents the velocity at the Eulerian point $\mathbf{x}_n = (x_n, y_n, z_n)$, h is the grid size of the Eulerian mesh, M is the number of Lagrangian points, and N is the number of Eulerian points affected by the immersed boundary. $D(\mathbf{x}_n - \mathbf{X}_m)$ is the Dirac delta function smoothly approximated by a continuous kernel distribution,¹⁸ and it is defined as follows:

$$D(\mathbf{x}_n - \mathbf{X}_m) = \frac{1}{h^3} \delta\left(\frac{x_n - X_m}{h}\right) \delta\left(\frac{y_n - Y_m}{h}\right) \delta\left(\frac{z_n - Z_m}{h}\right), \quad (9)$$

$$\delta(r) = \begin{cases} \frac{1}{8} \left(3 - 2|r| + \sqrt{1 + 4|r| - 4|r|^2} \right), & 0 \leq |r| \leq 1, \\ \frac{1}{8} \left(5 - 2|r| - \sqrt{-7 + 12|r| - 4|r|^2} \right), & 1 \leq |r| \leq 2, \\ 0, & 2 \leq |r|. \end{cases} \quad (10)$$

The Eulerian mesh area affected by the Lagrangian points is shown in Fig. 2. It can be seen that the interpolated region of each Lagrangian point is a square when the smooth Dirac delta function is used. Then, the velocity corrections of the affected Eulerian points are distributed from those of Lagrangian points through the Dirac delta function interpolation, which is written as

$$\delta \mathbf{u}(\mathbf{x}_n) = \sum_m \delta \mathbf{U}(\mathbf{X}_m) D(\mathbf{x}_n - \mathbf{X}_m) \Delta s_m, \quad (11)$$

where Δs_m is the area of the m th boundary element. Substituting Eqs. (7) and (11) into Eq. (8), we get

$$\mathbf{U}^{l+1}(\mathbf{X}_m) = \sum_n \mathbf{u}^*(\mathbf{x}_n) D(\mathbf{x}_n - \mathbf{X}_m) h^3 + \sum_n \sum_m \delta \mathbf{U}(\mathbf{X}_m) \Delta s_m D(\mathbf{x}_n - \mathbf{X}_m) D(\mathbf{x}_n - \mathbf{X}_m) h^3. \quad (12)$$

Equation (12) can be further written as a matrix form, that is

$$\mathbf{A} \mathbf{X} = \mathbf{B}, \quad (13)$$

where

$$\mathbf{X} = [\delta \mathbf{U}_1 \Delta s_1 \quad \delta \mathbf{U}_2 \Delta s_2 \quad \dots \quad \delta \mathbf{U}_M \Delta s_M]^T,$$

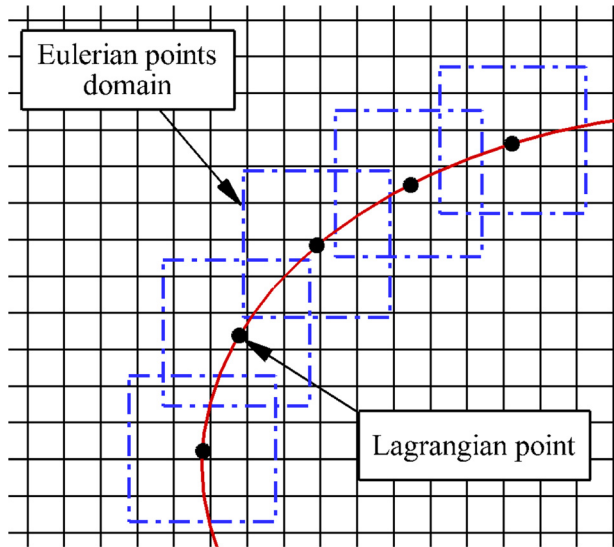


FIG. 2. Configuration of Lagrangian points and their surrounding Eulerian points (in two dimensions). The area enclosed by the blue dash dot line is the region where the fluid is affected by the Lagrangian point at the center. The area is a square when the smooth Dirac delta function is used for interpolation.

$$\mathbf{A} = \begin{bmatrix} D_{11} & D_{12} & \cdots & D_{1N} \\ D_{21} & D_{22} & \cdots & D_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ D_{M1} & D_{M2} & \cdots & D_{MN} \end{bmatrix} \begin{bmatrix} D_{11} & D_{12} & \cdots & D_{1M} \\ D_{21} & D_{22} & \cdots & D_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ D_{N1} & D_{N2} & \cdots & D_{NM} \end{bmatrix} h^3,$$

$$\mathbf{B} = \begin{bmatrix} \mathbf{U}_1 \\ \mathbf{U}_2 \\ \vdots \\ \mathbf{U}_M \end{bmatrix}^{t+1} - \begin{bmatrix} D_{11} & D_{12} & \cdots & D_{1N} \\ D_{21} & D_{22} & \cdots & D_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ D_{M1} & D_{M2} & \cdots & D_{MN} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^* \\ \mathbf{u}_2^* \\ \vdots \\ \mathbf{u}_N^* \end{bmatrix} h^3.$$

Actually, vector $\mathbf{B}$ denotes the velocity correction at the boundary (Lagrangian points), and matrix $\mathbf{A}$ is only related to the Lagrangian points and their surrounding Eulerian points, which can be precomputed. By solving the equation system (13) and using Eq. (11), we can obtain the velocity correction at the affected Eulerian points. Then, substituting it into Eq. (7), the corrected velocity field will be obtained.

III. INVERSE DISTANCE WEIGHTING INTERPOLATION-BASED IMMERSED BOUNDARY VELOCITY CORRECTION METHOD

A. Correction of the velocity field with inverse distance weighting method

The IDW method, originally proposed by Shepard,²¹ is an interpolation function for irregularly spaced data. It can be expressed as

$$\phi_o = \sum_k \gamma_k \phi_k / q, \quad \gamma_k = \left(\frac{1}{d_k} \right)^e, \quad q = \sum_k \gamma_k, \quad e > 0, \quad (k = 1, 2, \dots, K), \quad (14)$$

in which ϕ represents a generic flow variable, K is the number of the neighboring points associated with the interpolated point, and e is the power index of the inverse distance $1/d_k$. d_k is the distance between the interpolated point and the neighboring point, and it is calculated by

$$d_k = \sqrt{(x_k - x_o)^2 + (y_k - y_o)^2 + (z_k - z_o)^2}, \quad (15)$$

where (x_o, y_o, z_o) is the coordinate of the interpolated point, and (x_k, y_k, z_k) is the coordinate of the k th neighboring point associated with the interpolated point. Later, Franke³⁰ developed an alternative formulation of IDW, which is expressed as

$$\phi_o = \sum_k \gamma_k \phi_k / q, \quad \gamma_k = \left(\frac{R - d_k}{R d_k} \right)^e, \quad (16)$$

$$q = \sum_k \gamma_k, \quad e > 0, \quad (k = 1, 2, \dots, K),$$

where R represents a given interpolation radius, which can be set as the maximum value of distance d_k .

It can be found from Eqs. (14) and (16) that the selection of the interpolation range of IDW is very flexible. There are two ways to select nearby interpolation points. One is an arbitrary distance criterion (all data points within a radius R), and the other is an arbitrary number criterion (the nearest K data points).²¹ In the present work, we choose the arbitrary distance criterion to select nearby points as shown in Fig. 3. It can be seen that the interpolated region of each boundary point is a circle which is different from that of the smooth Dirac delta function. For convenience, Eqs. (14) and (16) can be rewritten as a unified form, that is

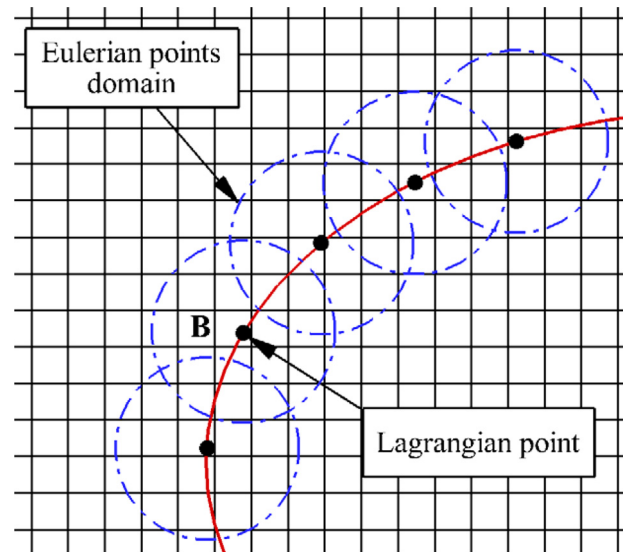


FIG. 3. Configuration of Lagrangian points and their surrounding Eulerian points (in two dimensions). The area enclosed by the blue dash dot line is the region where the fluid is affected by the Lagrangian point at the center. The area is a circular domain when the IDW method is used for interpolation.

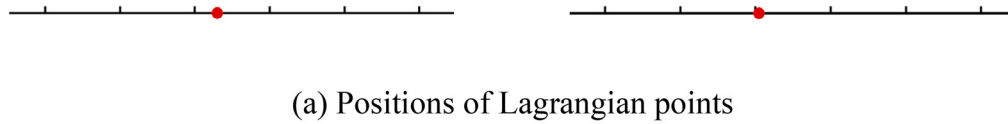
$$\begin{aligned}\phi_m &= \sum_k w_{mk} \phi_k = \sum_n w_{mn} \phi_n, \\ w_{mn} &= \begin{cases} w_{mk}, & d_{mn} \leq R, \\ 0, & d_{mn} > R, \end{cases} \\ w_{mk} &= \frac{\gamma_{mk}}{\sum_k \gamma_{mk}}, \quad d_{mk} \leq R, \end{aligned} \quad (17)$$

where m is the label of the interpolated point and n is the label of its neighboring points. In IBM, they represent the m th Lagrangian point

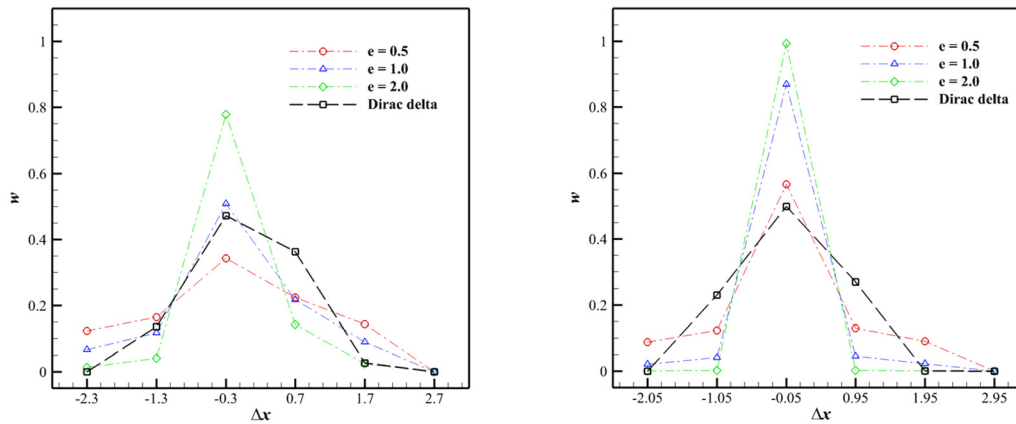
and the n th affected Eulerian points, respectively. d_{mn} is the distance between these two points.

In order to correct the velocity field, the flow variable ϕ is the velocity $\mathbf{u}$ here. Since the prediction step is the same as that in Sec. II B, we directly describe the procedure of the correction step. The velocity at the boundary point can be interpolated from the velocities of the neighboring points (take point B in Fig. 3 as an example), i.e.,

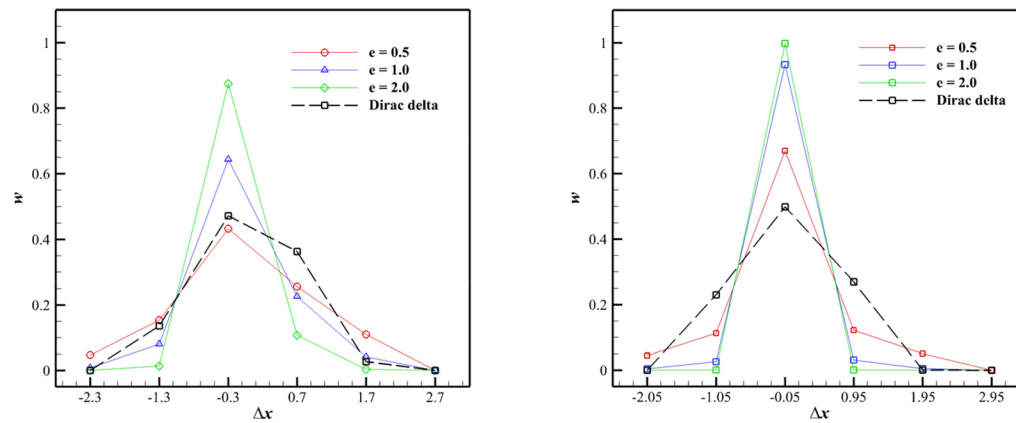
$$\mathbf{U}_{B,m}^{l+1} = \sum_n w_{mn} \mathbf{u}_n^{l+1} = \sum_k w_{mk} \mathbf{u}_k^{l+1}, \quad (m = 1, 2, \dots, M; \quad n = 1, 2, \dots, N; \quad k = 1, 2, \dots, K), \quad (18)$$



(a) Positions of Lagrangian points



(b) First IDW interpolation



(c) Second IDW interpolation

FIG. 4. Comparison of the distribution of weight coefficients in one dimension for two different IDW interpolation methods and the smooth Dirac delta function. (a) shows two different positions of the Lagrangian point, $x = -0.2$ and -0.45 from left to right (suppose that the length of the line segment is 1 and the position of the midpoint of the line segment is $x = 0$), (b) is the results of the first IDW interpolation proposed by Shepard,²¹ and (c) is the results of the second IDW interpolation proposed by Franke.³⁰

where $\mathbf{U}_{B,m}^{l+1}$ is the velocity at boundary point B. M and N are the same as those in Eq. (8), i.e., M is the number of the Lagrangian points and N is the number of Eulerian points affected by the immersed boundary. In addition, K represents the number of Eulerian points associated with the m th Lagrangian point B. Next, the velocity correction at the affected Eulerian points can be obtained by the following IDW interpolation, i.e.:

$$\delta \mathbf{u}_n = \sum_m w_{nm} \delta \mathbf{U}_m. \quad (19)$$

Same as Eq. (11), Eq. (19) means the velocity corrections of the n th affected Eulerian points are distributed from those of Lagrangian points, so w_{nm} is equal to w_{mn} in Eqs. (17) and (18). Substituting Eqs. (7) and (19) into Eq. (18), we have

$$\mathbf{U}_m^{l+1} = \sum_n w_{mn} \mathbf{u}_n^* + \sum_n \sum_m w_{mn} w_{nm} \delta \mathbf{U}_m. \quad (20)$$

The same as Eq. (12), Eq. (20) can also be further rewritten as a matrix form like Eq. (13), that is

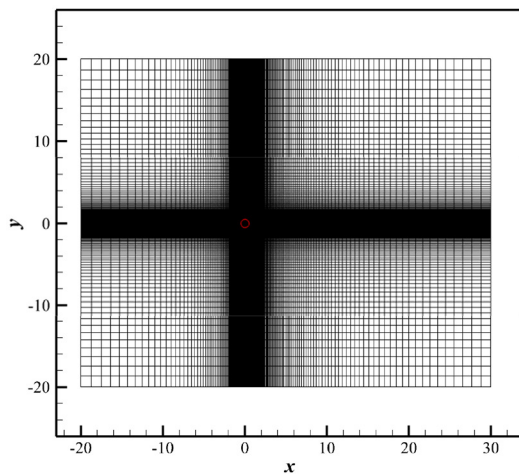
$$\mathbf{A}_{\text{IDW}} \mathbf{X}_{\text{IDW}} = \mathbf{B}_{\text{IDW}}, \quad (21)$$

where,

$$\mathbf{X}_{\text{IDW}} = [\delta \mathbf{U}_1 \quad \delta \mathbf{U}_2 \quad \cdots \quad \delta \mathbf{U}_M]^T,$$

$$\mathbf{A}_{\text{IDW}} = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1N} \\ w_{21} & w_{22} & \cdots & w_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ w_{M1} & w_{M2} & \cdots & w_{MN} \end{bmatrix} \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1M} \\ w_{21} & w_{22} & \cdots & w_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ w_{N1} & w_{N2} & \cdots & w_{NM} \end{bmatrix},$$

$$\mathbf{B}_{\text{IDW}} = \begin{bmatrix} \mathbf{U}_1 \\ \mathbf{U}_2 \\ \vdots \\ \mathbf{U}_M \end{bmatrix}^{l+1} - \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1N} \\ w_{21} & w_{22} & \cdots & w_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ w_{M1} & w_{M2} & \cdots & w_{MN} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^* \\ \mathbf{u}_2^* \\ \vdots \\ \mathbf{u}_N^* \end{bmatrix}.$$



Then, by solving the matrix system (21) and using Eq. (19), the velocity correction at the affected Eulerian points can be calculated. Next, according to Eq. (7), the corrected velocity field will be obtained.

It should be emphasized that it is quite convenient to calculate the force on the immersed boundary in the current method. According to Eq. (5), the force density at the affected Eulerian points is

$$\mathbf{f} = \rho \frac{\delta \mathbf{u}_n}{\delta t}. \quad (22)$$

From Newton's third law, it can be known that the lift and drag of the flow field acting on the immersed boundary are, respectively,

$$F_L = - \sum_n \frac{\rho \delta v_n S_n}{\delta t}, \quad (23)$$

$$F_D = - \sum_n \frac{\rho \delta u_n S_n}{\delta t}, \quad (24)$$

where S_n is the area (in two dimensions) or the volume (in three dimensions) of the n th affected Eulerian cell.

B. Power index of inverse distance

In Sec. III A, two forms of the IDW interpolation [Eqs. (14) and (16)] have been shown. In this section, we will compare the distribution of weight coefficients in one dimension for both IDW interpolation methods and the smooth Dirac delta function, and then determine the suitable parameter ϵ , which is the power index of the inverse distance.

Since the positions of the Lagrangian points in the Eulerian mesh are random, two representative cases are selected for comparison. As shown in Fig. 4(a), the red solid point represents the Lagrangian point and the black line segments represent connections between Eulerian points. Assuming that the length of the line segment is 1 and the position of the midpoint of the line segment is $x = 0$, the positions of the Lagrangian points in these two cases are, respectively, $x = -0.2$ and -0.45 . The weight coefficients of the IDW interpolation [Eq. (14)]

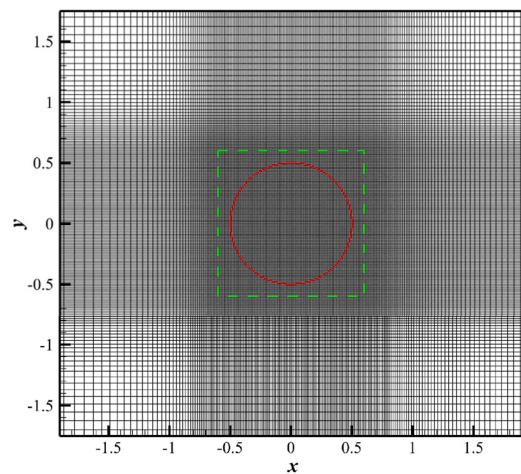


FIG. 5. Computational domain (left) and local enlarged view of the uniform mesh region (right) for the flow past a circular cylinder. The red solid line is the immersed boundary and the green dotted box represents the uniform mesh region.

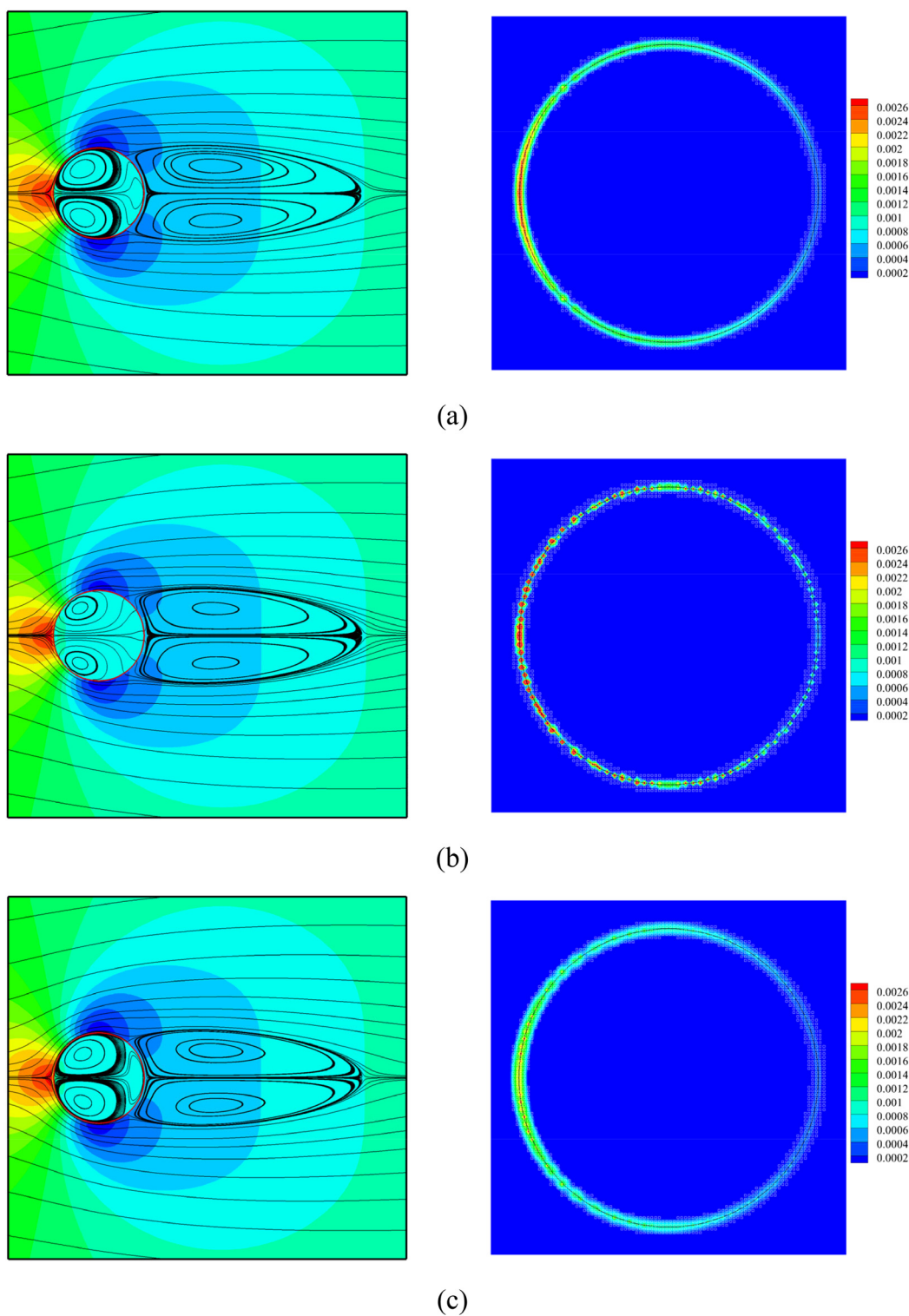


FIG. 6. Streamlines (left) and distributions of the velocity corrections (right) for the flow past a circular cylinder at $Re = 40$ by using different interpolation methods. The number of Lagrangian points is 120. (a) The smooth Dirac delta function; (b) the IDW interpolation with the power index $e = 2.0$; and (c) the IDW interpolation with the power index $e = 0.5$.

proposed by Shepard²¹ are shown in Fig. 4(b) and the weight coefficients of the IDW interpolation [Eq. (16)] proposed by Franke³⁰ are shown in Fig. 4(c). In Figs. 4(b) and 4(c), three values of the parameter e are compared, which are respectively $e = 0.5$, 1.0, and 2.0. Here, the radius of the IDW interpolation is set as $R = 2.5h$.

From Fig. 4, it can be seen that when $e = 1.0$ and 2.0, the distribution of the weight coefficients of the IDW interpolation deviates significantly from those of the Dirac delta function. The weight coefficients of points closest to the Lagrangian point are too large, and the weight coefficients of other points are too small, so the influence of the immersed boundary may not be well transferred to the flow field. Hence, setting $e = 0.5$ is more appropriate. The detailed comparison will be shown in Sec. IV A. In addition, when $e = 0.5$, the IDW interpolation [Eq. (16)] proposed by Franke³⁰ is better according to our numerical experiment. Therefore, we mainly use this IDW method in the present work.

IV. NUMERICAL EXAMPLES

In this section, some numerical examples will be tested to verify the present method and the results will be compared with the reference data. All the examples are for incompressible flows. In the prediction step, the predicted flow field is given by the lattice Boltzmann flow solver (LBFS).³⁶

A. Flow past a circular cylinder

The flow over a circular cylinder is chosen to validate the proposed method. This problem has been extensively studied, and many theoretical, experimental, and numerical results are available in the literature. Different types of flow behavior for this problem can be characterized by the Reynolds number, which is defined as

$$Re = \frac{\rho U_{\infty} D}{\mu}, \quad (25)$$

where $D = 1$ is the diameter of the circular cylinder, and U_{∞} is the reference velocity at infinity.

First, the steady flow at $Re = 40$ is considered to compare the effect of the power index of inverse distance on the flow field in IDW interpolation. The computational domain is set to $[-20D, 20D] \times [-20D, 30D]$, and the whole mesh size is 301×251 . As shown in Fig. 5, the center of the cylinder is $(0, 0)$, and the cylinder is located in a $1.2D \times 1.2D$ uniform mesh area with a mesh size of 97×97 . So the minimum mesh spacing for the uniform mesh is $h = 0.0125$ and the radius of the IDW interpolation is set to $R = 2.5h$. The cylinder is represented by 120 Lagrangian points with uniform distribution.

Figure 6 shows streamlines around the cylinder and the velocity correction distributions obtained by different interpolation methods. In the right figures, the black solid line represents the immersed boundary and the white hollow squares denote the Eulerian points associated with the Lagrangian points through interpolation functions. Figure 6(a) is the result of the smooth Dirac delta function, and Figs. 6(b) and 6(c) are the results of the IDW interpolation with different power indices of inverse distance. It can be seen clearly that there is no streamline penetration to the cylindrical surface in Figs. 6(a) and 6(c). However, in Fig. 6(b), when the power index of inverse distance is $e = 2.0$, some streamlines pass through the cylinder. Comparing the distribution of the velocity corrections, in the results of

the smooth Dirac delta function and the IDW interpolation with the power index $e = 0.5$, there is a clear diffusion effect on the cylindrical surface, thereby transferring the influence of the immersed boundary to the surrounding flow field. However, for the case of the power index $e = 2.0$, the influence of the immersed boundary is not well transferred to the fluid points, since the velocity corrections are concentrated on the Eulerian points closest to the Lagrangian points. This phenomenon is consistent with the judgment in Sec. III B. Figure 7 combines the streamlines and the distribution of the velocity corrections for the case of IDW with the power index $e = 2.0$. It can be seen clearly that the discontinuity of the boundary diffusion effect transmitted from adjacent Lagrangian points to the flow field is the reason why the streamline penetrates the surface of the cylinder. This situation gets worse when the number of Lagrangian points decreases.

Furthermore, the number of Lagrangian points is reduced to 80 and the same test is repeated. Here, we only compare the results obtained by using the smooth Dirac delta function and the IDW interpolation with the power index $e = 0.5$ as shown in Fig. 8. In Fig. 8(a), it can be found that the smooth Dirac delta function can no longer guarantee that the streamlines do not penetrate the cylindrical surface, and the distribution of the velocity corrections also shows the discontinuity of the boundary diffusion effect transferred from adjacent Lagrangian points to the flow field. Figure 9 shows clearly that the streamlines pass through the location where the distribution of the velocity corrections is discontinuous. However, Fig. 8(b) shows that there is still no streamline passing through the cylinder when the IDW interpolation is used.

In addition, Table I shows more results for different numbers of Lagrangian points. From Table I, it can be concluded that the IDW

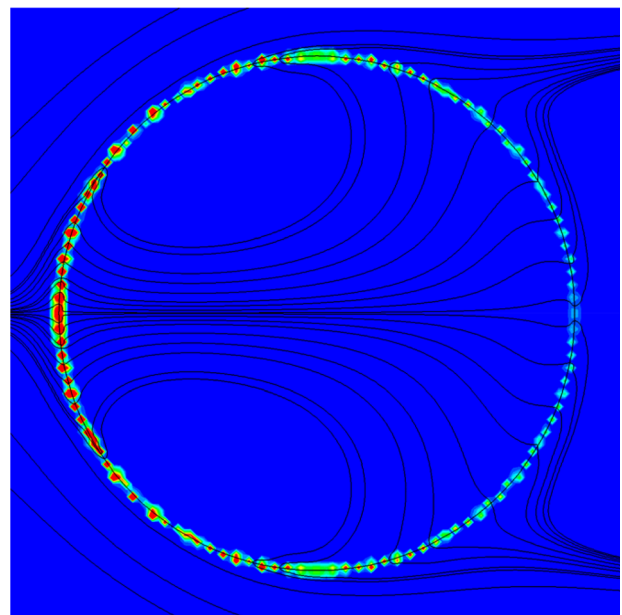


FIG. 7. Streamlines and distributions of the velocity corrections for the flow past a circular cylinder at $Re = 40$ by using the IDW method with the power index $e = 2.0$. The number of Lagrangian points is 120.

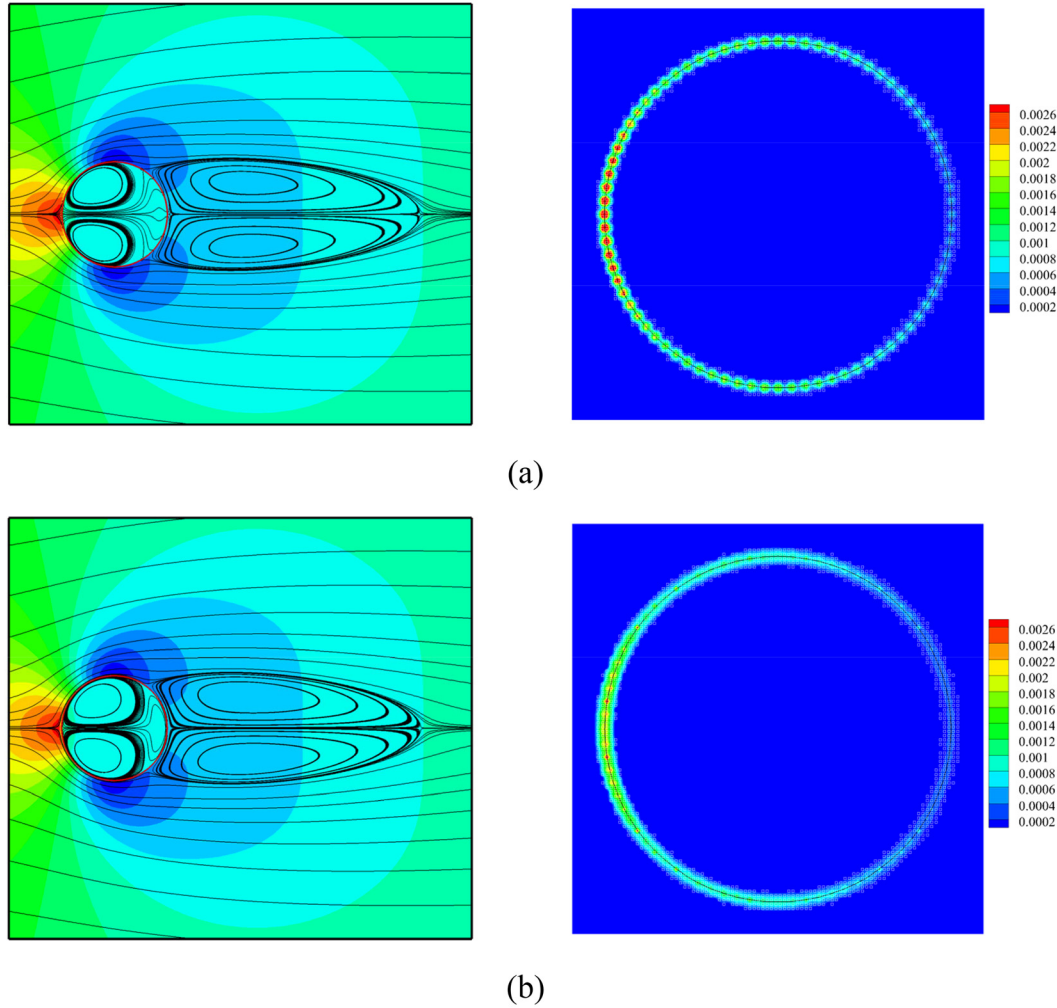


FIG. 8. Streamlines (left) and distributions of the velocity corrections (right) for the flow past a circular cylinder at $Re = 40$ by using different interpolation methods. The number of Lagrangian points is 80. (a) The smooth Dirac delta function and (b) the IDW interpolation with the power index $e = 0.5$.

interpolation with the power index $e = 0.5$ used in the present method allows a larger ratio of mesh size between the Lagrangian grid points and Eulerian mesh than the smooth Dirac delta function. This means that for the same Eulerian mesh, we can use fewer Lagrangian points which improves the computational efficiency.

Furthermore, the steady flow at $Re = 20$ with the same Eulerian mesh is tested. To save computational time, the number of Lagrangian points is taken as 80. Table II lists the drag coefficient C_d , recirculating length $2L/D$ and separation angle θ_s obtained by using the present method with IDW and they are compared with Refs. 27 and 37–40. The drag coefficient C_d is defined as

$$C_d = \frac{F_D}{\frac{1}{2} \rho U_\infty^2 D}, \quad (26)$$

in which F_D is the drag force calculated by Eq. (24). Both the results of the steady flows at $Re = 20$ and 40 are presented in Table II. It can be

found that the numerical results are in good agreement with those in the literature. In addition, the velocity contours are plotted in Fig. 10.

To further validate the present method, the unsteady flow around the cylinder at $Re = 100$ and 200 are simulated. In this case, the flow is in a periodical state, and there will be alternating shedding of vortex. The settings of the computational domain and the Eulerian mesh are all the same as for the steady flow case, except that the whole mesh size is set to 501×351 . The instantaneous vorticity contours and the streamlines are plotted in Figs. 11 and 12. The Karman vortex street behind the cylinder can be clearly seen in these figures. Figure 13 presents the time evolution of drag and lift coefficients. They both reveal distinct periodicity. Tables III and IV compare the calculated results of drag coefficient C_d , lift coefficient C_l and Strouhal number St with previous numerical results. Here, the drag coefficient C_d is time-averaged. Similar to the drag coefficient, the lift coefficient C_l is computed by

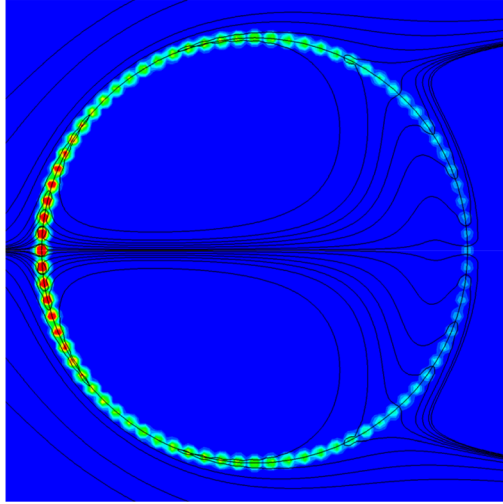


FIG. 9. Streamlines and distributions of the velocity corrections for the flow past a circular cylinder at $Re = 40$ by using the smooth Dirac delta function. The number of Lagrangian points is 80.

TABLE I. Comparison of streamlines penetrating the cylinder by using different interpolation methods with different ratios of mesh sizes (minimum Eulerian mesh spacing is $h = 0.0125$) for flow around a circular cylinder at $Re = 40$.

Interpolation methods	Number of Lagrangian points	Boundary element arc length Δs	Ratio of mesh sizes $\Delta s/h$	Streamlines penetrating the cylindrical surface
Dirac delta function	120	$\pi/120$	2.0944	No
IDW with $e = 2.0$				Yes
IDW with $e = 0.5$				No
Dirac delta function	100	$\pi/100$	2.5133	No
IDW with $e = 0.5$				No
Dirac delta function	90	$\pi/90$	2.7925	Yes
IDW with $e = 0.5$				No
Dirac delta function	80	$\pi/80$	3.1416	Yes
IDW with $e = 0.5$				No
Dirac delta function	70	$\pi/70$	3.5904	Yes
IDW with $e = 0.5$				Yes

$$C_l = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 D}, \quad (27)$$

where F_L is the lift force calculated by Eq. (23). The Strouhal number is defined as

$$St = \frac{f_q D}{U_\infty}, \quad (28)$$

TABLE II. Comparison of drag coefficient C_d , recirculating length $2L/D$, and separation angle θ_s for steady flow over a circular cylinder at $Re = 20$ and 40 . The present method uses the IDW interpolation with the power index of inverse distance $e = 0.5$.

Reynolds numbers	References	C_d	$2L/D$	θ_s ($^\circ$)
20	Dennis and Chang ³⁷	2.045	1.88	43.7
	Russell and Wang ³⁸	2.13	1.88	43.3
	Choi <i>et al.</i> ²⁷	2.02	1.80	40.8
	Taira and Colonius ³⁹	2.06	1.88	43.3
	Kolahdouz <i>et al.</i> ⁴⁰	2.10	1.86	44.4
	Present	2.078	1.88	41.8
40	Dennis and Chang ³⁷	1.522	4.69	53.8
	Russell and Wang ³⁸	1.60	4.58	53.1
	Choi <i>et al.</i> ²⁷	1.52	4.50	51.0
	Taira and Colonius ³⁹	1.54	4.60	53.7
	Kolahdouz <i>et al.</i> ⁴⁰	1.58	4.62	54.1
	Present	1.555	4.76	51.8

where f_q is the shedding frequency of vortex. Tables III and IV show that our results are in good agreement with those of Refs. 8 and 40–44.

B. Numerical test of overall accuracy

In the present work, the LBFS, which has the second-order accuracy, is adopted to calculate the basic flow field. The IDW interpolation with the power $e = 0.5$ is used to get the velocity correction at Eulerian points. Although it only works in the region near the immersed boundary, it may have an impact on the accuracy of the solution in the whole computational domain. Therefore, we numerically analyze the overall accuracy of the present method in this section. The decaying vortex problem⁴⁵ is chosen to access the accuracy of this method. The analytic solution to this problem is available, which can be expressed as

$$u(x, y, t) = -U \cos\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) e^{\frac{-2\pi^2 U t}{ReL}}, \quad (29)$$

$$v(x, y, t) = U \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi y}{L}\right) e^{\frac{-2\pi^2 U t}{ReL}}, \quad (30)$$

$$\rho(x, y, t) = \rho_0 - \frac{\rho_0 U^2}{4C_s^2} \left[\cos\left(\frac{2\pi x}{L}\right) + \cos\left(\frac{2\pi y}{L}\right) \right] e^{\frac{-4\pi^2 U t}{ReL}}. \quad (31)$$

In this test, the computational domain is set to $[-L, L] \times [-L, L]$, in which $L = 1$. U is the characteristic velocity and the Reynolds number is $Re = \rho UL/\mu = 10$. A circular cylinder with the radius of 0.5 is immersed in the flow field and the exact solution is imposed along the immersed boundary. The exact solution is also used to provide the initial and boundary conditions. When the numerical solutions at time $Ut/L = 1$ are obtained, the numerical error of u can be quantified by using L_2 norm, which is defined as

$$L_2 \text{ error} = \sqrt{\frac{\sum \left(\frac{u_{\text{numerical}} - u_{\text{exact}}}{U} \right)^2}{NE}}, \quad (32)$$

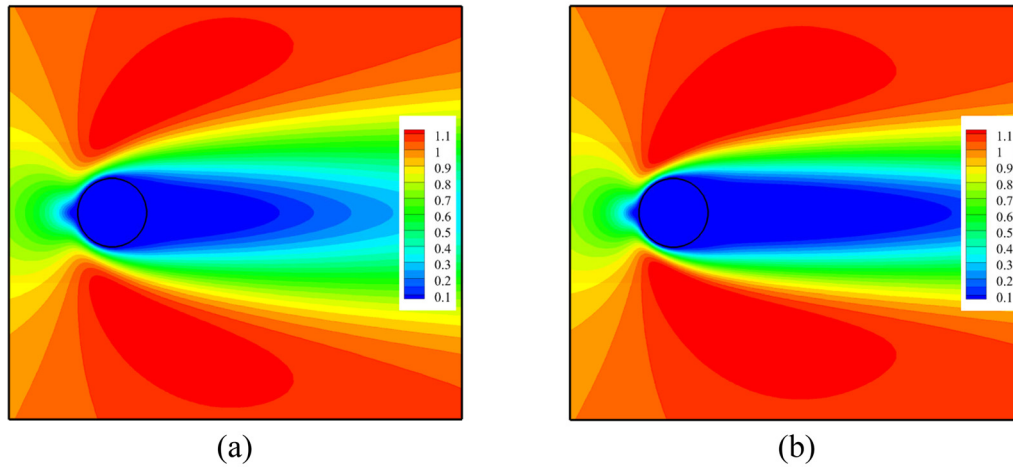
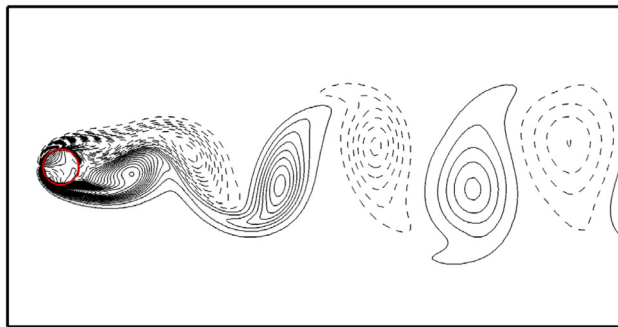
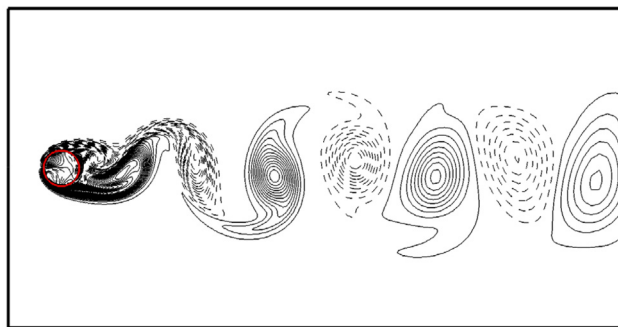


FIG. 10. Velocity contours for flow over a cylinder at (a) $Re = 20$ and (b) $Re = 40$.



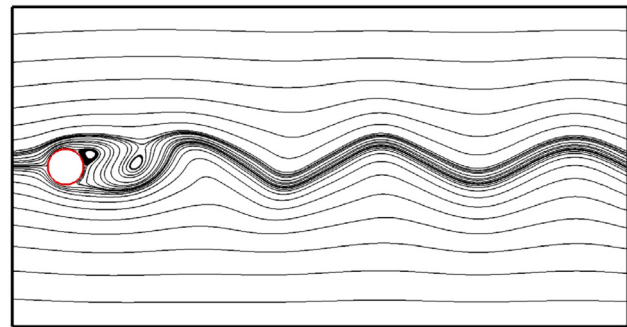
(a)



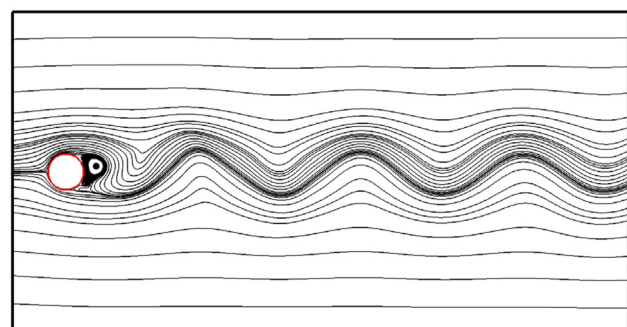
(b)

FIG. 11. Vorticity contours for flow over a cylinder at (a) $Re = 100$ and (b) $Re = 200$. The dotted and solid lines denote the negative and positive levels of vorticity, respectively.

where $u_{\text{numerical}}$ and u_{exact} are the numerical and exact results, respectively. $NE = (N-1) \times (N-1)$ is the total number of Eulerian points in the whole domain. Five different uniform meshes ($N \times N$, $N = 11, 21, 41, 81$, and 161) are used in the simulations and the characteristic



(a)



(b)

FIG. 12. Streamlines for flow over a cylinder at (a) $Re = 100$ and (b) $Re = 200$.

velocity U is correspondingly set to $0.1, 0.05, 0.025, 0.0125$, and 0.00625 . Table V lists the local and overall L_2 errors of the velocity component u , where the local errors record only the Eulerian points affected by the Lagrangian points. In addition, the maximum error of

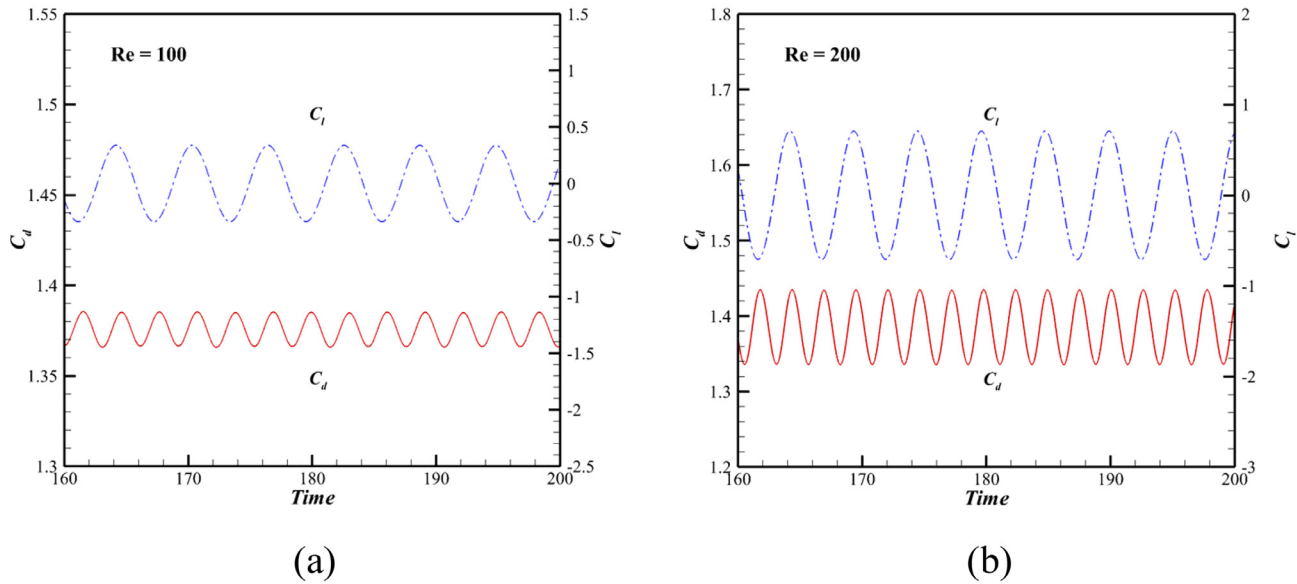


FIG. 13. Evolution of drag and lift coefficients for flow over a cylinder at (a) $Re = 100$ and (b) $Re = 200$.

TABLE III. Comparison of drag coefficient C_d , lift coefficient C_l and Strouhal number St for unsteady flow over a cylinder at $Re = 100$. (The value of C_d is time-averaged.).

References	C_d (averaged)	C_l	St
Liu <i>et al.</i> ⁴⁴	1.35	± 0.339	0.165
Wu and Shu ⁸	1.364	± 0.344	0.163
Kolahdouz <i>et al.</i> ⁴⁰	1.37	± 0.351	0.168
Zhao <i>et al.</i> ⁴¹	1.36	± 0.32	0.167
Yan <i>et al.</i> ⁴²	1.387	± 0.340	0.166
Present	1.376	± 0.338	0.163

TABLE IV. Comparison of drag coefficient C_d , lift coefficient C_l and Strouhal number St for unsteady flow over a cylinder at $Re = 200$. (The value of C_d is time-averaged.).

References	C_d (averaged)	C_l	St
Liu <i>et al.</i> ⁴⁴	1.31	± 0.69	0.192
Kolahdouz <i>et al.</i> ⁴⁰	1.39	± 0.75	0.198
Zhao <i>et al.</i> ⁴¹	1.42	± 0.75	0.192
Yan <i>et al.</i> ⁴²	1.372	± 0.691	0.197
Barbeau <i>et al.</i> ⁴³	1.395	± 0.71	0.200
Present	1.385	± 0.709	0.194

the velocity component u is also shown in Table V. The rate of convergence is calculated by taking the logarithm for the ratio of two successive errors as

$$\text{Rate} = \log_2 \frac{E_h}{E_{h/2}}, \quad (33)$$

where E_h denotes the error when the mesh size is h . The numerical error vs mesh spacing in the log scale is presented in Fig. 14. As shown in Table V and Fig. 14, both the overall accuracy and local accuracy of numerical results for the present method are slightly lower than the second order. It can be concluded that the impact of this IBM on the computational accuracy is relatively small.

C. Flow over a NACA0012 airfoil

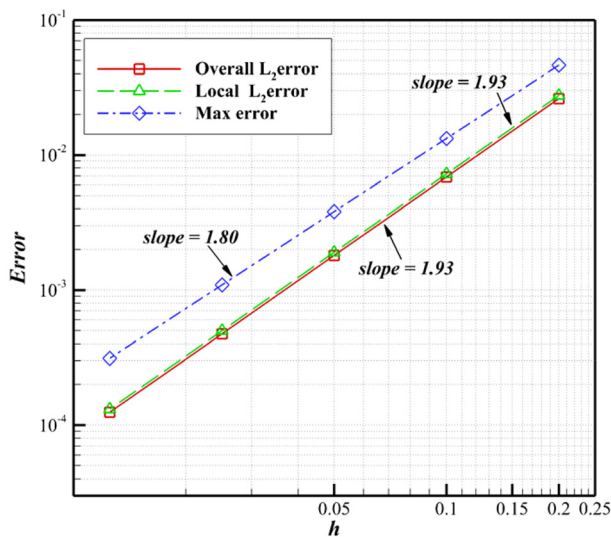
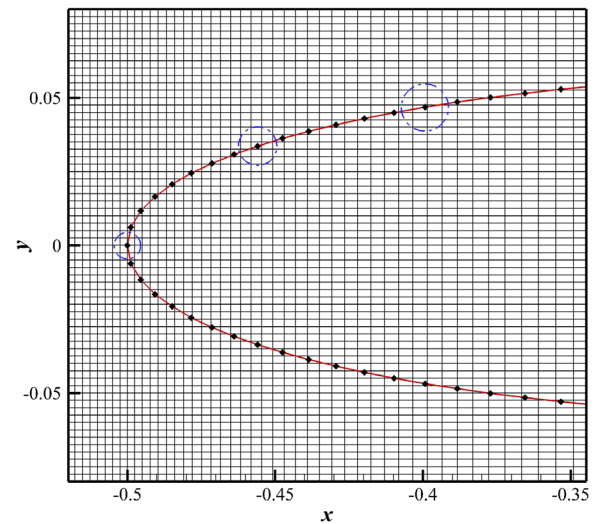
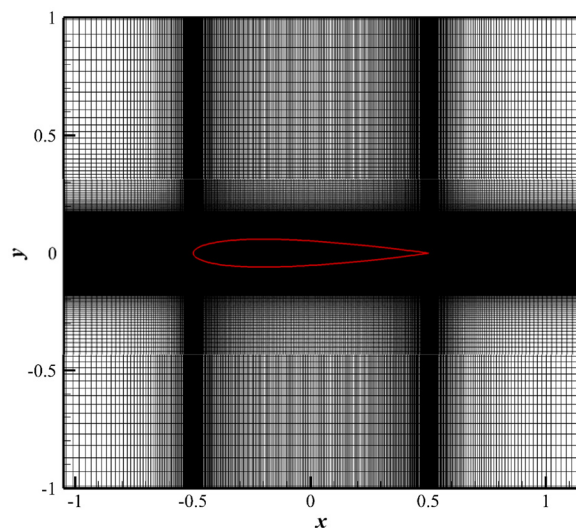
In this section, the flow over a NACA0012 airfoil will be simulated by using the proposed method. In the previous sections, the circular cylinder is always represented by Lagrangian points with uniform distribution. However, for an airfoil, more points need to be distributed on the leading and trailing edges, which means that the total number of Lagrangian points increases if the uniform distribution is adopted. In order to reduce the number of Lagrangian points, the airfoil is represented by points with the non-uniform distribution. Correspondingly, the Eulerian mesh near the airfoil can also be set non-uniformly. As shown in Fig. 15, there are more points near the leading and trailing edges of the airfoil, so the Eulerian mesh is denser in these regions. In this non-uniform mesh, the smooth Dirac delta function is no longer applicable. However, the IDW interpolation can take advantage of the flexibility of its interpolation range. Here, we associate the interpolation range of IDW with the distribution of Lagrangian points, and let the interpolation range vary with the arc length Δs of the boundary element as shown in Fig. 16. The interpolation range is defined as

$$R_m = \alpha \Delta s_m, \quad (34)$$

where R_m is the interpolation radius of the m th Lagrangian point, and α is a scaling factor. The setting of α can refer to the case of flow past a circular cylinder in Sec. IV A, where the interpolation radius of IDW is $R = 2.5h$ (here $h = 0.0125$) and the number of Lagrangian points is

TABLE V. L_2 error of the velocity component u for decaying vortex flow.

Mesh sizes	Number of Lagrangian points	Maximum error	Local L_2 error	Overall L_2 error	Rate (Overall)
11×11	6	4.753×10^{-2}	2.663×10^{-2}	2.473×10^{-2}	...
21×21	12	1.436×10^{-2}	7.746×10^{-3}	7.259×10^{-3}	1.77
41×41	24	3.384×10^{-3}	1.887×10^{-3}	1.860×10^{-3}	1.96
81×81	48	9.977×10^{-4}	4.880×10^{-4}	4.773×10^{-4}	1.96
161×161	96	3.484×10^{-4}	1.327×10^{-4}	1.206×10^{-4}	1.98

**FIG. 14.** Convergence of numerical error vs mesh spacing for decaying vortex problem.**FIG. 16.** Schematic diagram of the variation of IDW interpolation range with the arc length of boundary element. The black solid points are Lagrangian points and the circles of blue dash dot line are interpolation range of IDW.**FIG. 15.** Local enlarged view of the computational domain for flow over a NACA0012 airfoil. The red solid line is the airfoil.

$M = 80$ – 120 . Thus, the corresponding value of α for the case of flow past a circular cylinder can be calculated by

$$\alpha = \frac{R}{\Delta s} = \frac{2.5hM}{\pi} \approx 0.8\text{--}1.2. \quad (35)$$

In the test case of flow around a NACA0012 airfoil, α is set to 0.8.

At first, the flow around the NACA0012 airfoil at $Re = 500$ and the zero angle of attack is tested. The Reynolds number is defined as $Re = \rho U_\infty c / \mu$, where the airfoil chord length $c = 1$. The airfoil is represented by 160 Lagrangian points with non-uniform distribution and the minimum Lagrangian mesh spacing near the leading and trailing edges is 0.00625. The computational domain is set to $[-15c, 15c] \times [-20c, 30c]$, and the whole mesh size is 501×301 . The minimum Eulerian mesh spacing near the leading and trailing edges of the airfoil is $h = 0.0025$. For this test, the drag and pressure coefficients are computed by

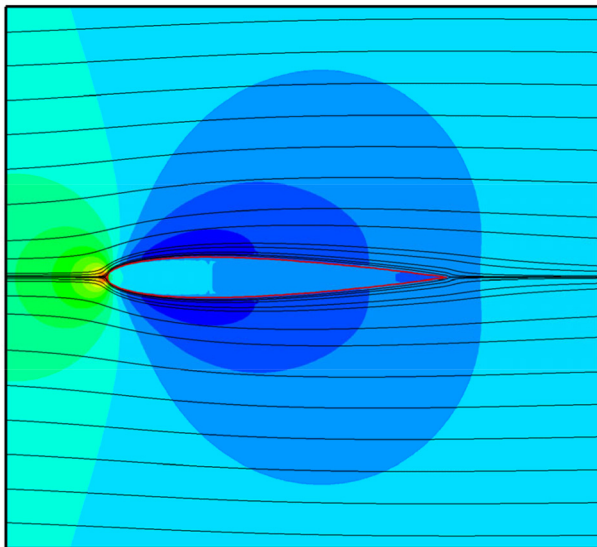
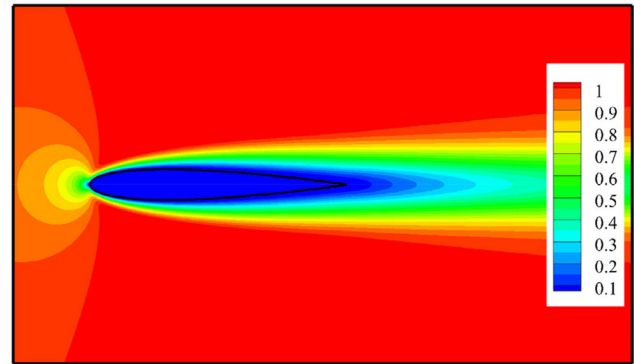
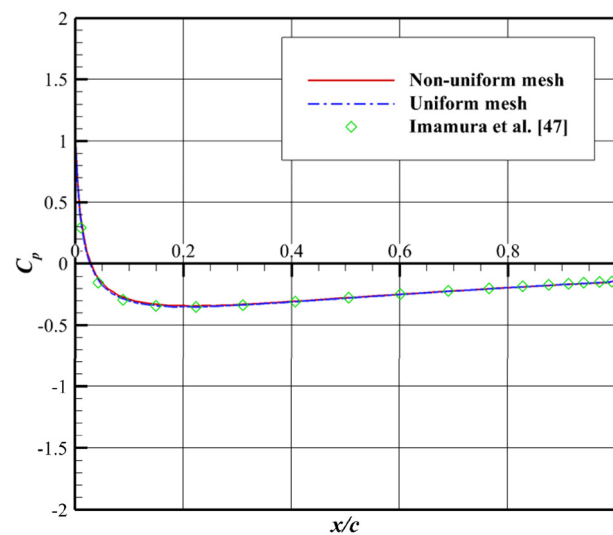
$$C_d = \frac{F_D}{\frac{1}{2} \rho U_\infty^2 c}, \quad (36)$$

TABLE VI. Comparison of drag coefficient C_d for flow over a NACA0012 airfoil at $Re = 500$.

References	C_d
CFL3D ⁴⁶	0.1762
Imamura <i>et al.</i> ⁴⁷	0.1736
Wu and Shu ⁸	0.1759
Present	0.1759

$$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U_\infty^2}, \quad (37)$$

where F_D is the drag force calculated by Eq. (24) and p_∞ is the reference pressure. The comparison of drag coefficients C_d obtained by the present method with Refs. 8, 46, and 47 is listed in Table VI. It can be seen that the obtained drag coefficient C_d is in agreement with the reference data. Figure 17 shows the pressure contours together with the streamlines and Fig. 18 shows the velocity contours. Meanwhile, the same test is carried out by the smooth Dirac delta function. Due to the use of the Dirac delta function, the Eulerian mesh is distributed uniformly around the airfoil, and the NACA0012 airfoil is also represented by Lagrangian points with uniform distribution. In order to maintain the mesh resolution of the leading and trailing edges of the airfoil, the number of Lagrangian points and Eulerian points must increase. Thus, the airfoil is represented by 320 Lagrangian points and the whole mesh size is 681×301 . The pressure coefficient distributions along the airfoil surface are presented in Fig. 19, and the velocity profiles of the airfoil at five cross sections are shown in Fig. 20. The results obtained by the present method with the non-uniform mesh

**FIG. 17.** Pressure contours and streamlines for flow over NACA0012 airfoil at $Re = 500$ and the zero angle of attack.**FIG. 18.** Velocity contours for flow over NACA0012 airfoil at $Re = 500$ and the zero angle of attack.**FIG. 19.** Distribution of pressure coefficient C_p along the NACA0012 airfoil at $Re = 500$ and the zero angle of attack.

are basically consistent with the results obtained by the smooth Dirac delta function-based-IBM with uniform mesh, as well as the numerical results of Ref. 47. What is more, Table VII shows the CPU time required by different methods. It can be seen that the present method takes less time due to the use of fewer mesh points. Therefore, it can be concluded that the present method with the non-uniform mesh has higher computational efficiency on the premise of ensuring the correct calculation results.

In addition, in order to verify the ability of the present method to simulate moving boundary problems, the flow over a heaving NACA0012 airfoil is tested. The motion of the airfoil is given by the following equation:

$$Y = 0.25t^2(3 - t), \quad (38)$$

where Y is the vertical displacement of the airfoil. The time for the heaving motion is $t = 0-2$, during which the airfoil moves from $Y = 0$

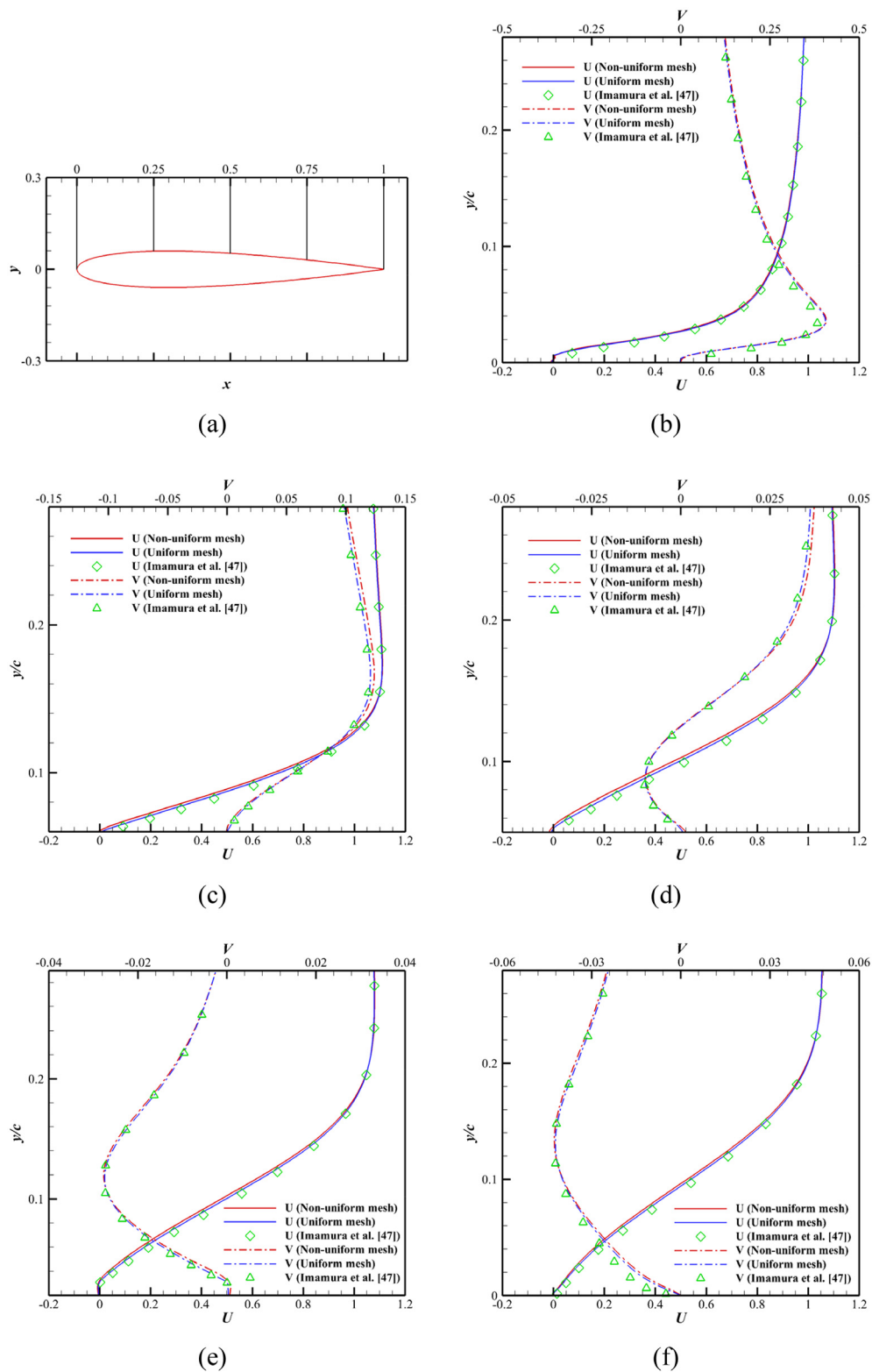


FIG. 20. Velocity profiles of the NACA0012 airfoil at five cross sections. (a) Cross sections, (b) $x = 0.00$, (c) $x = 0.25$, (d) $x = 0.50$, (e) $x = 0.75$, and (f) $x = 1.00$.

TABLE VII. CPU time required by different methods for the stationary NACA0012 airfoil cases.

Methods	CPU Time (h)
Non-uniform mesh with the IDW method (501 × 301)	8.314
Uniform mesh with the smooth Dirac delta function (681 × 301)	10.578

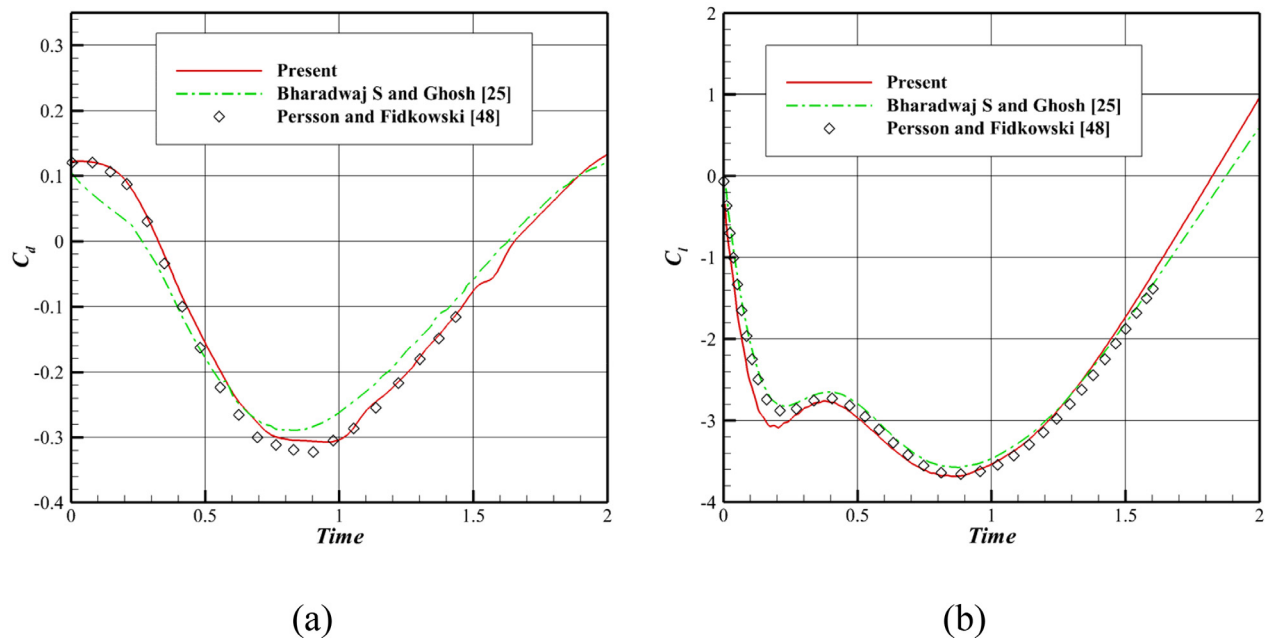
to $Y = 1$. The Reynolds number is set to 1000. Same as the previous example, the airfoil is represented by 160 Lagrangian points with non-uniform distribution and the Eulerian mesh is denser near the leading and trailing edges. The computational domain is still set to $[-15c, 15c] \times [-20c, 30c]$. Since the airfoil moves in the y -direction, the Eulerian mesh points should be increased in the y -direction. Thus, the whole mesh size is 501×691 . The time histories of lift and drag coefficients (C_l and C_d) are plotted in Fig. 21. It can be seen that both two force coefficients compare well with the literature.^{25,48} Furthermore, the airfoil represented by 320 Lagrangian points with uniform distribution is also tested. The Eulerian mesh with uniform distribution around the airfoil is used, and the whole mesh size is 681×691 . Figure 22 shows the pressure coefficient C_p along the upper and lower surfaces of the airfoil at different times and the vorticity contours at the same time instants. The pressure coefficient C_p obtained by using the uniform mesh with the smooth Dirac delta function is also presented in Fig. 22. It can be seen that the pressure coefficient C_p predicted by the present method with non-uniform mesh is consistent with the reference data²⁵ and the result obtained

by using the uniform mesh. This example proves that the present method can well simulate the moving boundary problem in non-uniform meshes. In addition, Fig. 23 shows the CPU time required by the present method with non-uniform mesh and the smooth Dirac delta function-based IBM with uniform mesh. Obviously, the present method improves computational efficiency since it uses fewer mesh points.

D. Flow past a stationary sphere

To further illustrate the capability of the present method for the three-dimensional simulation, the flow past a stationary sphere is tested. Since the IDW interpolation is based on the distance between points, its formula is the same in two dimensions and three dimensions. Therefore, the proposed method can be easily extended to the three-dimensional case. In the simulation, the computational domain is taken as $[-10D, 15D] \times [-10D, 10D] \times [-10D, 10D]$ in the x -, y -, and z -directions, where $D = 1$ is the diameter of the sphere. The whole mesh size is $111 \times 101 \times 101$. The sphere is located in a $1.2D \times 1.2D \times 1.2D$ rectangular box with a uniform mesh size of $49 \times 49 \times 49$, resulting in the minimum mesh spacing for the uniform mesh to be $h = 0.025$. The sphere is discretized by using 1357 Lagrangian points and its center is $(0, 0, 0)$.

The Reynolds number of this problem is defined as $Re = \rho U_\infty D / \mu$. Two cases of $Re = 100$ and 250 are considered. As reported in the previous study,¹⁴ for the incompressible flow past a stationary sphere, the flow regimes are steady axisymmetric flow at $Re = 100$ and steady non-axisymmetric flow at $Re = 250$. These flow regimes can be clearly observed in Fig. 24, which shows streamlines in x - y and x - z planes. The velocity contours for the steady axisymmetric flow at $Re = 100$ are plotted in Fig. 25. In addition, Table VIII lists the

**FIG. 21.** Time history of force coefficients for flow past a heaving NACA0012 airfoil at $Re = 1000$. (a) Drag coefficient C_d and (b) lift coefficient C_l .

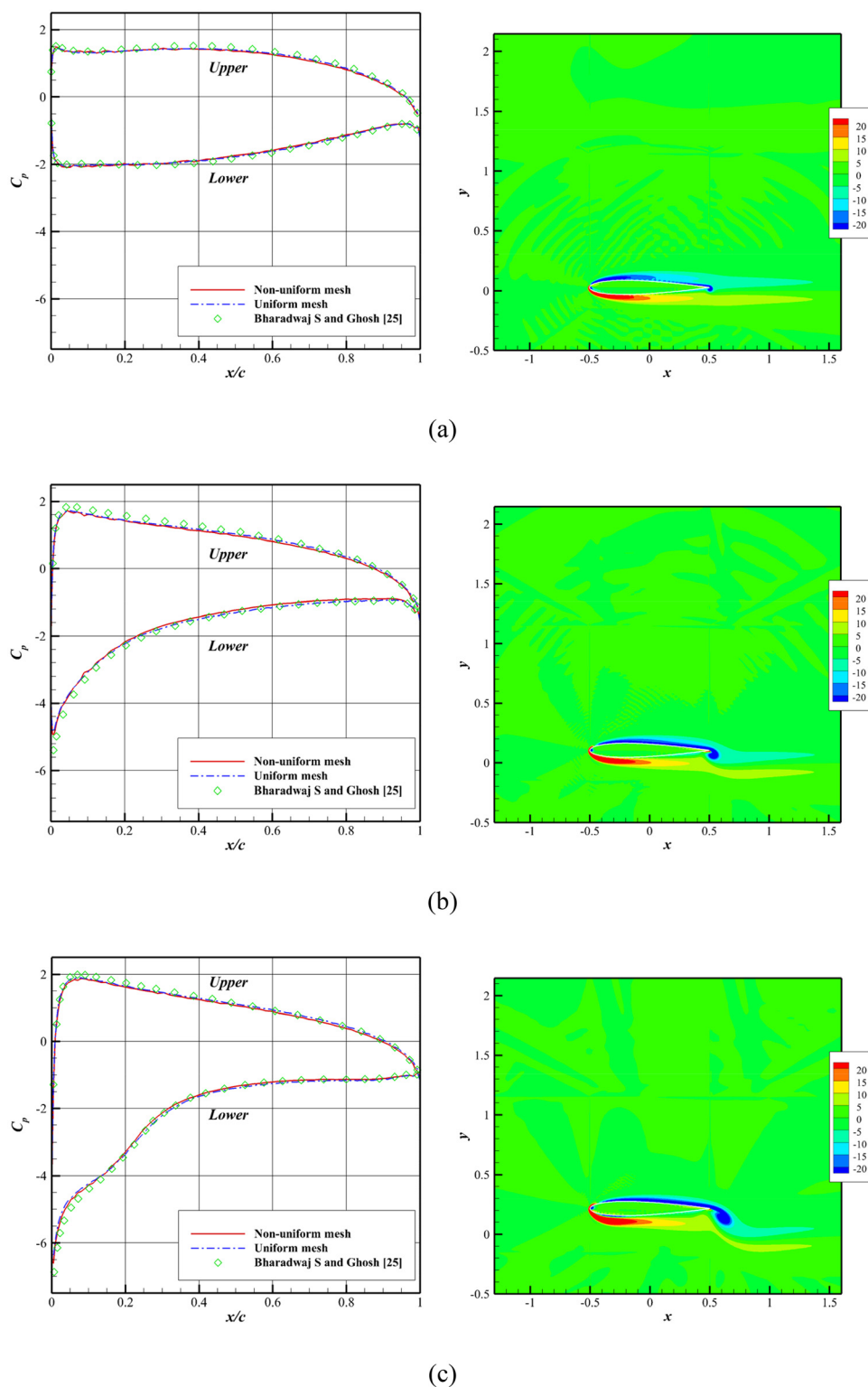
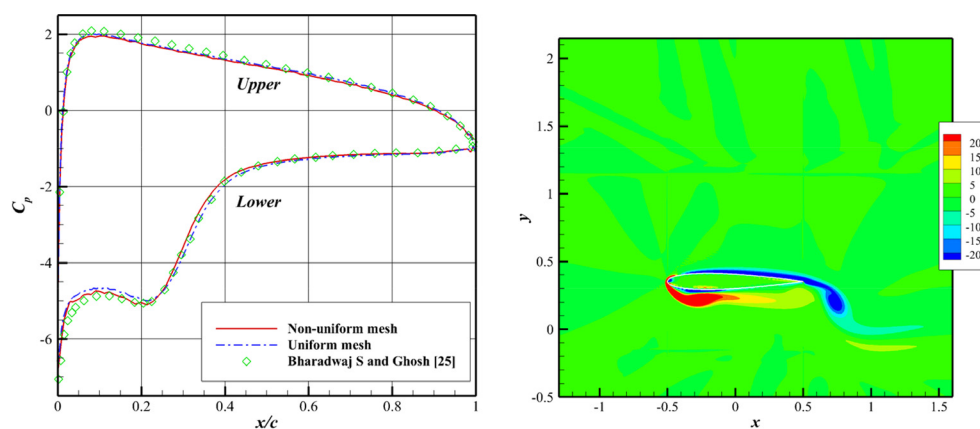
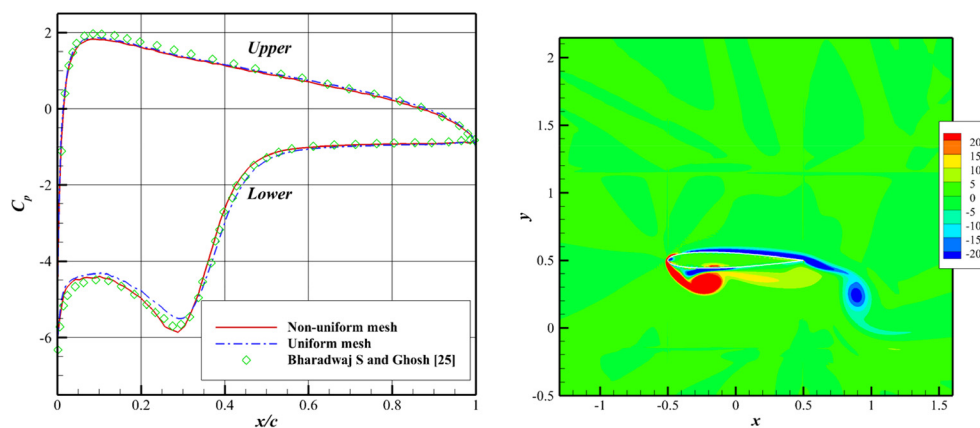


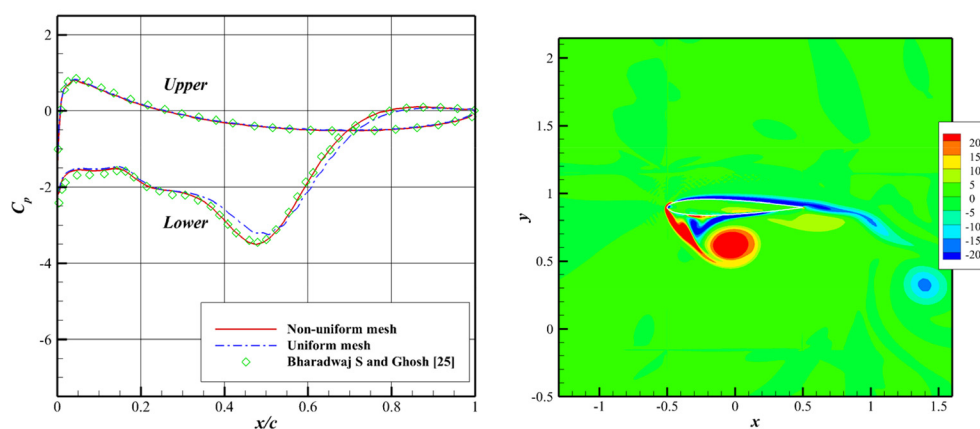
FIG. 22. Distribution of the pressure coefficient C_p (left) and vorticity contours (right) at different times for flow past a heaving NACA0012 airfoil at $Re = 1000$. (a) $t = 0.2$, (b) $t = 0.4$, (c) $t = 0.6$, (d) $t = 0.8$, (e) $t = 1.0$, (f) $t = 1.6$, and (g) $t = 2.0$.



(d)



(e)



(f)

FIG. 22. (Continued.).

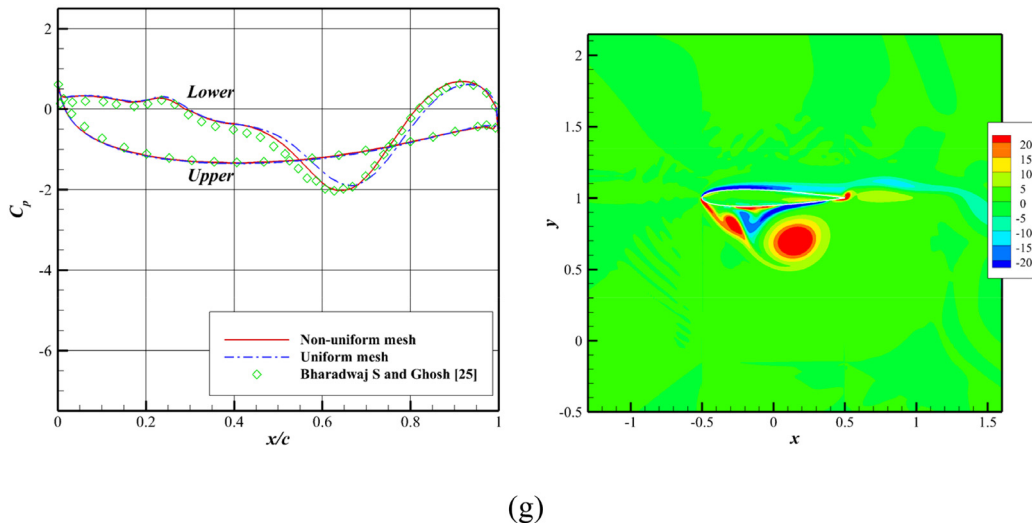


FIG. 22. (Continued.).

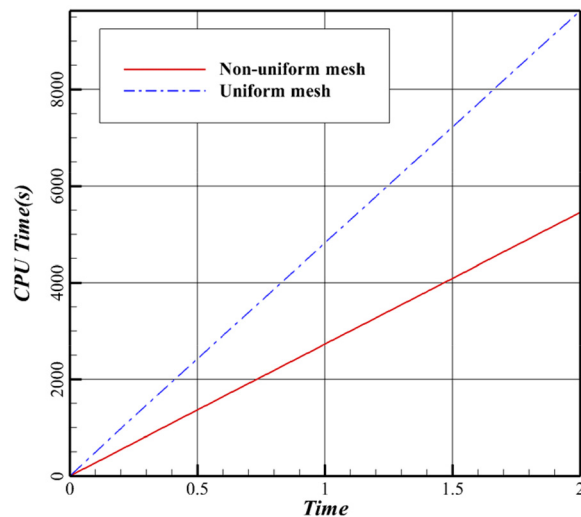


FIG. 23. CPU time required by the present method with non-uniform mesh and the smooth Dirac delta function-based IBM with uniform mesh for the moving NACA0012 airfoil case.

drag coefficient C_d for the flow at $Re = 100$ and 250 . Here, the drag coefficient C_d is defined as

$$C_d = \frac{F_D}{\frac{1}{2} \rho U_\infty^2 \pi r^2} = \frac{8F_D}{\rho U_\infty^2 \pi D^2}, \quad (39)$$

where F_D is the drag force calculated by Eq. (24). It can be seen from Table VIII that the present results agree well with the reference data.^{15,49–51} Obviously, the proposed method can effectively simulate the three-dimensional flow.

V. CONCLUSIONS

The implicit velocity correction-based IBM developed by Wu and Shu⁸ is a diffuse-interface IBM, which enforces the no-slip boundary condition by solving an equation system. Since the no-slip boundary condition can be guaranteed strictly, it avoids the unphysical phenomenon of the conventional diffuse-interface IBM that streamlines may pass through the solid body. However, in this IBM, the smooth Dirac delta function is used to associate Lagrangian points and their surrounding Eulerian points, limiting its application in uniform meshes. The IDW method originally proposed by Shepard²¹ is a typical spatial interpolation method, which is very flexible in selecting the interpolation range. By using the IDW interpolation, the diffuse-interface IBM can get rid of the limitation of the smooth Dirac delta function. This work presents an IDW interpolation-based immersed boundary velocity correction method for simulation of incompressible flows.

After comparing the coefficient distribution of the smooth Dirac delta function and the IDW function, we suggest a suitable power index of inverse distance. In the numerical experiments, the advantage of IDW to flexibly set the interpolation range is fully exploited. For steady and unsteady flows past a circular cylinder, a suitable interpolation range help to reduce the number of Lagrangian points while ensuring streamline does not penetrate the solid body and calculating force coefficients accurately. Then, the interpolation range is set to vary with the non-uniform distribution of Lagrangian points in the numerical experiment for flow around a NACA0012 airfoil. This makes the proposed method be used in a non-uniform mesh, which can improve the computational efficiency due to the use of fewer mesh points. In addition, the numerical test of the decaying vortex problem proves that the IDW used in our method does not significantly affect the overall computational accuracy. Finally, the simulation of the flow past a stationary sphere demonstrates that the proposed method can effectively simulate the three-dimensional flow.

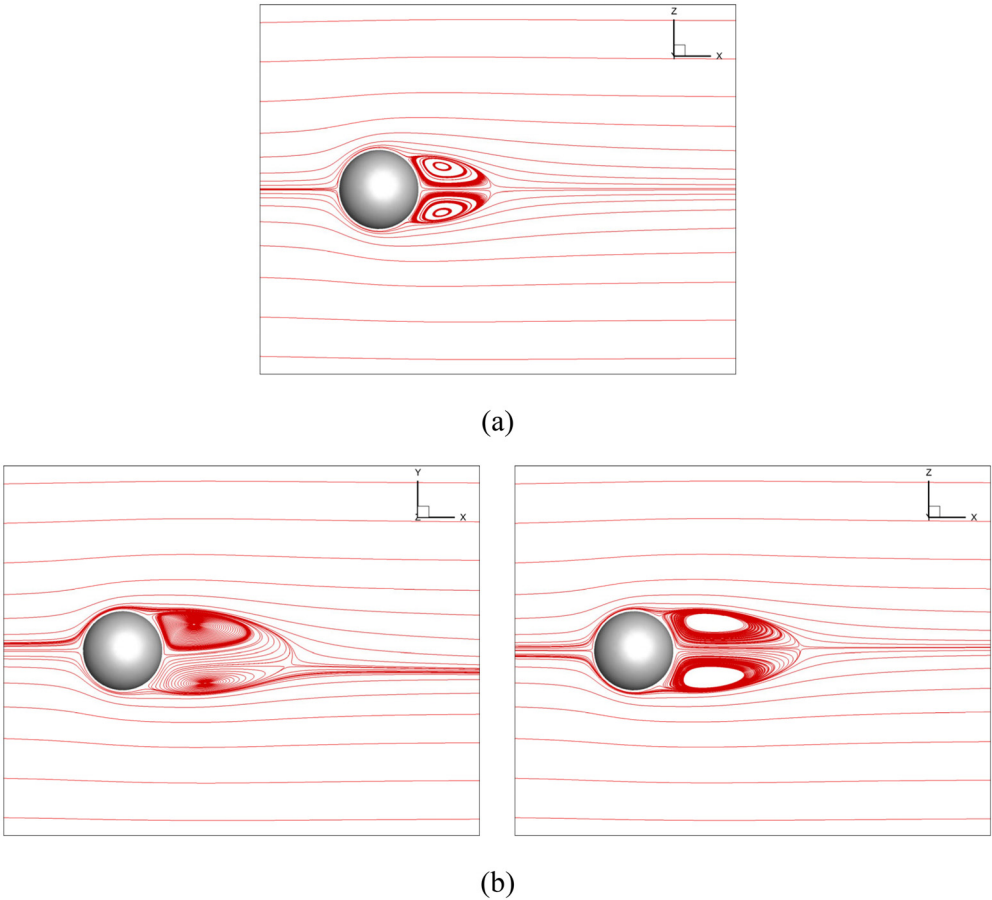


FIG. 24. Streamlines for flow past a stationary sphere. (a) Steady axisymmetric flow at $Re = 100$ in x - z plane and (b) steady non-axisymmetric flow at $Re = 250$ in x - y (left) and x - z (right) planes.

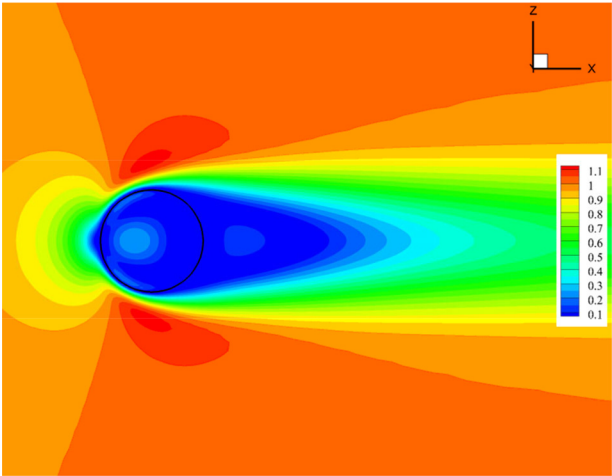


FIG. 25. Velocity contours for flow past a stationary sphere at $Re = 100$.

TABLE VIII. Comparison of the drag coefficient C_d for a flow past a stationary sphere at $Re = 100$ and 250 .

References	$Re = 100$	$Re = 250$
Gilmanov <i>et al.</i> ⁴⁹	1.153	...
Wang <i>et al.</i> ⁵⁰	1.108	0.746
Yang <i>et al.</i> ¹⁵	1.130	0.723
Chen <i>et al.</i> ⁵¹	1.148	...
Present	1.143	0.720

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yinjie Du: Conceptualization (equal); Data curation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Liming Yang:** Conceptualization (equal); Formal analysis (equal); Supervision (equal); Writing – review & editing (equal). **Chang Shu:** Conceptualization (equal); Formal analysis (equal); Investigation (lead); Supervision (equal); Writing – review & editing (equal). **Yang Xiao:** Data curation (supporting). **Yuxin Song:** Investigation (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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